

Rural water point siting in Masindi District, Uganda

SUBJECT	Placement of 10 additional water points under a fixed budget
PREPARED BY	An autonomous analysis agent, from open data, without field verification
DECISIONS	All 7 decisions in this run were recorded by a person.
REVIEWED BY	Not reviewed. No planner overrides were applied to this run.
GENERATED	2026-08-27T03:06:57Z
RUN	masindi-water-20260827T030535Z
PROVENANCE	0e193d1063ade2ac

Statement

An estimated 816,304 people live in the area of interest. 264,334 of them, 32.4%, live within about 14 minutes of walking over the modelled terrain of a water point recorded as serving. 551,970 people, 67.6% of the district, do not (Ex01, Ex02, Ex03).

Of the 2,178 water point records held for this district, 1,752 are recorded as serving and 426 are not (Ex01). The median record is 4.5 years old, so the figures above describe the surveyed state of the network rather than its state today (Ex10).

14,563 candidate locations were evaluated (Ex04). Selecting 10 of them by greedy coverage maximisation brings a further 34,700 people inside the service radius, raising coverage to 36.6% (Ex05, Ex06). Returns fall away quickly: the first site reaches 12,223 people and the last 54, so the number of sites to build is a decision that should be taken on the curve at Ex05 rather than on a round number.

No planner overrides were applied. The recommendation at Ex06 is the unmodified output of the selection method at Ex05.

3 of the 10 independent checks at Ex08 returned a flag rather than a pass: boundary effect, equity, cartographic consistency. The flagged findings are reproduced in full at Ex08 and are not resolved in this bundle.

3 properties of the source registers would have changed a coverage figure had they been handled differently. Each is recorded at Ex10 with what was observed and what was done about it. None were corrected silently.

Recommendations in this bundle are advisory. Land availability, tenure, water table, community consent and construction feasibility are outside the data used here and inside the judgement of the officer reviewing it. No figure in this bundle is asserted without an exhibit number after it.

Index of exhibits

Ex	Content	What it supports
Ex00	Decisions taken, and by whom	That the judgements this assessment rests on are attributable
Ex01	Rural water point siting register for Masindi	The installed stock and its recorded condition
Ex02	Population surface	The denominator for every coverage figure in this bundle
Ex03	Coverage before any intervention	The size of the gap the recommendation is trying to close
Ex04	Candidate sites considered	That the recommendation was selected from a stated, reproducible set
Ex05	Selection method and its guarantee	That the ranking is reproducible and its optimality is bounded
Ex06	Recommended sites	The recommendation itself, in coordinates
Ex07	Planner overrides and what they cost	That the recommendation was reviewed by a person, and where it was overruled
Ex07b	Safety gate decisions	That access, transfer and fallback decisions were taken deliberately
Ex08	Independent checks	That the plan was checked by a process that did not produce it
Ex08b	Scoring review	That the account was scored against a stated floor before issue
Ex09	Sensitivity of the recommendation	How far the recommendation depends on assumptions the planner can change
Ex10	Anomalies in the source data, and how they were handled	That the source registers were read rather than assumed
Ex11	Reproduction record	That this bundle can be regenerated from the recorded queries

Verification

Every retrieval in this bundle is recorded with its endpoint, the query as executed, the time of retrieval and the record counts before and after cleaning (Ex01, Ex02, Ex11). Cleaning rules are declared per source in the handbook files that accompany this bundle and are applied in order, each reporting its own drop count, so no record leaves the pipeline unaccounted for.

Coverage is recomputed independently of the solver by a separate process before this bundle is produced, and that process can prevent it being produced at all (Ex08). Coverage is always the union over serving locations; a per-location sum double counts every settlement reachable from two of them, and Ex08 reports what that sum would have claimed.

Properties of the source data that would change a figure if read differently are recorded at Ex10 rather than handled silently. Where the agent's own source handbook was found to be incomplete, that is recorded there too.

No figure in this bundle was estimated, interpolated or supplied from memory. A figure is either computed from a recorded retrieval or it is absent.

Ex00 – Decisions taken, and by whom

SUPPORTS CAPTURED METHOD

That the judgements this assessment rests on are attributable

Recorded before the analysis ran

Each row is a judgement about values rather than a technical parameter: what counts as served, whose need the programme answers, whether a register of this age is fit to direct spending. The analysis refuses to run without them rather than assuming one, because assuming one would move the decision away from the officer accountable for it. Where a row is attributed to the agent, no district position was recorded and the agent decided in its absence.

All 7 decisions in this run were recorded by a person.

Decision	Value	Recorded by	Reason given
service_radius_m	1000 metres	District Water Officer, Masindi	One kilometre corresponds to the 30 minute round trip the JMP uses as a basic drinking water service threshold, and to the figure the district already reports against. Adopted for continuity with existing reporting rather than because the terrain has been assessed.
coverage_basis	walking_time	District Water Officer	Reach is measured as walking time over the modelled terrain rather than as straight-line distance, because the straight-line measure counts households as served across the Kafu and the seasonal swamps that they cannot in fact cross on foot. The friction surface is coarse and does not know local footpaths, and that limitation is accepted in preference to ignoring terrain altogether.
objective	worst_case	District Water Officer, Masindi	The 2026 district development plan prioritises the refugee settlement sub-counties and the dispersed eastern parishes, which are the households currently furthest from any serving point. Reducing the worst-served distance is therefore the objective, accepting a lower total covered than maximum coverage would reach. The analyst reported the trade as roughly 12,000 people of coverage against a fall in the worst-served distance from 31.5 km to 8.1 km.
budget	10 facilities	District Water Officer, Masindi	Ten boreholes are funded under the FY2026 rural water capital allocation.
data_currency_accepted	yes, with the age stated in the report	District Water Officer, Masindi	The register's median record is 4.4 years old, ranging from 3.5 to 16.2 years. Accepted as the best available basis for a siting decision that cannot wait for a resurvey, on the condition that the assessment states the age prominently and describes its findings as the surveyed state of the network rather than its present state.
coverage_tolerance	0.05 share of achievable coverage	District Water Officer, Masindi	Local vetoes forgoing more than five per cent of achievable coverage are to be reported to the District Water and Sanitation Coordination Committee rather than settled at officer level.
equity_accepted	yes	District Water Officer, Masindi	With the worst_case objective the plan is directed at remote households by design, so the distributional concern that applies to coverage maximisation does not arise in the same form here. The measured distribution is accepted.

Ex01 – Rural water point siting register for Masindi

SUPPORTS CAPTURED METHOD

The installed stock and its recorded condition

<https://data.waterpointdata.org/resource/eqje-vguj.json> on 2026-08-27

Socrata query below, executed verbatim. Cleaning rules are declared in handbooks/wpdx.yaml and applied in order; each rule reports its own drop count.

```
clean_country_name = 'Uganda' AND clean_adm2 = 'Masindi'
```

Records returned	2,189
Records after cleaning	2,178
Discarded	11

RECORDS DISCARDED, BY RULE

flagged duplicate by WPdx	11
Recorded as serving	1,752
Recorded as not serving	426
Median record age	4.5 years

Licence: CC BY 4.0

Ex02 – Population surface

SUPPORTS	The denominator for every coverage figure in this bundle
CAPTURED	https://hub.worldpop.org/rest/data/pop/wpgp on 2026-08-27
METHOD	National 100 m grid, read over the area of interest and aggregated into blocks. The aggregation is coarser than the source and finer than the service radius, so it does not bias coverage.

```
UGA ppp 2020, bbox 31.3199 1.3260 32.3629 2.3721 (WGS 84), aggregated 5x5
```

Raster cells in window	62,750
Aggregated demand cells	60,638
Population in the area of interest	816,304

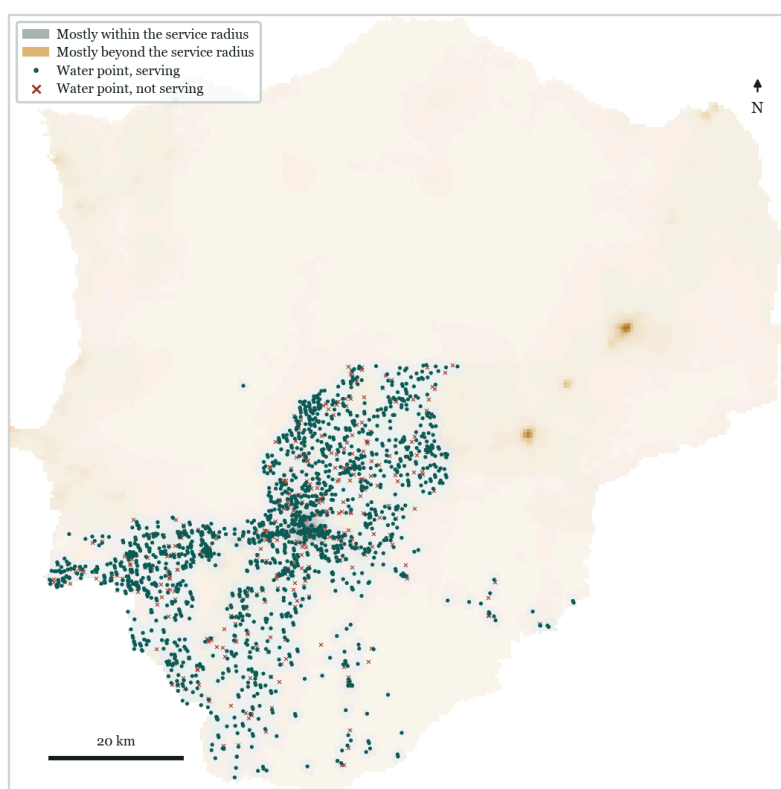
population year 2020; 1,571,260 source cells at ~100 m read over the window, aggregated 5x5 into 62,750 cells of ~500 m

Licence: CC BY 4.0

Ex03 – Coverage before any intervention

SUPPORTS	The size of the gap the recommendation is trying to close
CAPTURED	Computed from Ex01 and Ex02
METHOD	A demand cell counts as covered when it lies within about 14 minutes of walking over the modelled terrain of a point recorded as serving. Coverage is the union over serving points; a per-point sum would double count every settlement reachable from two of them.

Population	816,304
Covered today	264,334 (32.4%)
Not covered	551,970 (67.6%)



Serving and non-serving points over the population surface. Warm shading is population beyond the service radius.

Ex04 – Candidate sites considered

SUPPORTS	That the recommendation was selected from a stated, reproducible set
CAPTURED	Constructed from Ex01 and the area of interest
METHOD	Two kinds of candidate: an existing point recorded as not serving, which could be rehabilitated, and a regular lattice of greenfield locations. Lattice points already inside an existing service area are removed before selection.

Candidates evaluated	14,563
Lattice spacing	750 m
Minimum spacing between recommendations	1000 m

Ex05 – Selection method and its guarantee

SUPPORTS	That the ranking is reproducible and its optimality is bounded
CAPTURED	Computed
METHOD	Candidates are scored by the population they would newly bring inside the service radius and selected greedily. On a monotone submodular coverage function greedy selection is within a factor of $1 - 1/e$, about 63 per cent, of the optimum. The exact integer programme is not used in the loop: it takes minutes at district scale where greedy takes seconds, and every planner override triggers a re-solve.

Sites selected	10
Population newly covered	34,700
Coverage after the programme	36.6%
Benchmark against an exact solver	not run (not run)

Sites built	Newly covered	Cumulative	Share of district
1	12,223	276,557	33.9%
2	3,038	279,594	34.2%

Sites built	Newly covered	Cumulative	Share of district
3	8,865	288,460	35.3%
4	220	288,680	35.4%
5	2,000	290,680	35.6%
6	525	291,205	35.7%
7	2,393	293,597	36.0%
8	1,434	295,032	36.1%
9	3,949	298,980	36.6%
10	54	299,034	36.6%

Marginal and cumulative population covered. The first site reaches 12,223 people, the last 54.

Ex06 – Recommended sites

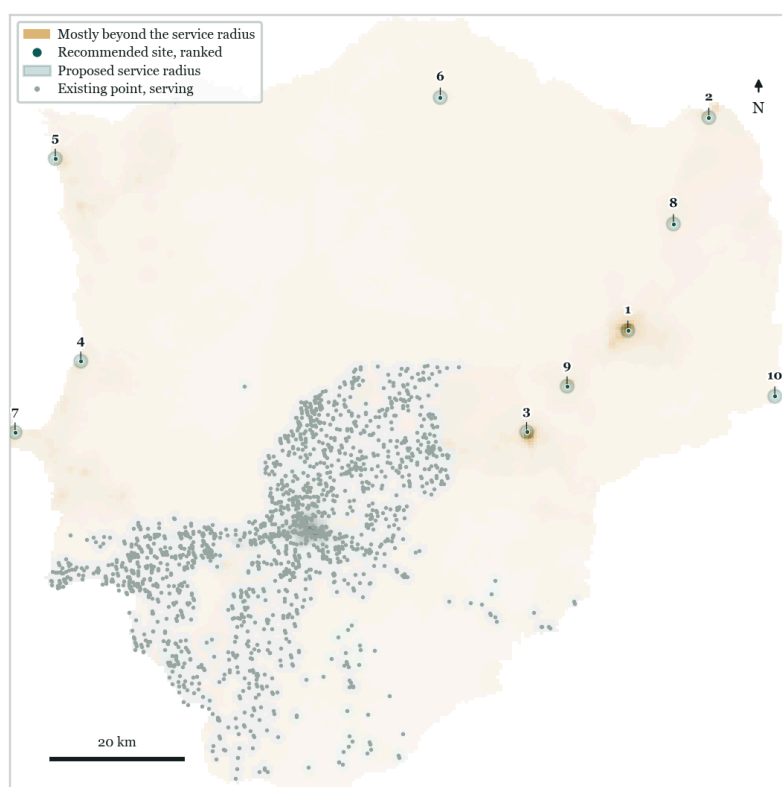
SUPPORTS
CAPTURED
METHOD

The recommendation itself, in coordinates

Computed

Sites in selection order. Each row's coverage is additional to every row above it, so the programme must be built from the top down. Coordinates are WGS 84.

No.	Site	Type	Location (WGS 84)	Newly covered	Nearest serving
1	S-001	greenfield	1.9501, 32.1419	12,223	25.6 km
2	S-002	greenfield	2.2351, 32.2497	3,038	52.3 km
3	S-003	greenfield	1.8144, 32.0071	8,865	12.1 km
4	S-004	greenfield	1.9089, 31.4138	220	24.5 km
5	S-005	greenfield	2.1802, 31.3798	2,000	19.6 km
6	S-006	greenfield	2.2620, 31.8922	525	38.9 km
7	S-007	greenfield	1.8139, 31.3263	2,393	17.2 km
8	S-008	greenfield	2.0926, 32.2026	1,434	38.2 km
9	S-009	greenfield	1.8754, 32.0611	3,949	16.3 km
10	S-010	greenfield	1.8620, 32.3375	54	42.5 km



Recommended sites and their service radii over the unserved population.

BASIS FOR EACH SITE

S-001	New site covering 12,223 people currently beyond the service radius; nearest working point 25.6 km away.
S-002	New site covering 3,038 people currently beyond the service radius; nearest working point 52.3 km away.
S-003	New site covering 8,865 people currently beyond the service radius; nearest working point 12.1 km away.
S-004	New site covering 220 people currently beyond the service radius; nearest working point 24.5 km away.
S-005	New site covering 2,000 people currently beyond the service radius; nearest working point 19.6 km away.
S-006	New site covering 525 people currently beyond the service radius; nearest working point 38.9 km away.
S-007	New site covering 2,393 people currently beyond the service radius; nearest working point 17.2 km away.
S-008	New site covering 1,434 people currently beyond the service radius; nearest working point 38.2 km away.
S-009	New site covering 3,949 people currently beyond the service radius; nearest working point 16.3 km away.
S-010	New site covering 54 people currently beyond the service radius; nearest working point 42.5 km away.

Ex07 – Planner overrides and what they cost

SUPPORTS CAPTURED METHOD	That the recommendation was reviewed by a person, and where it was overruled overrides YAML, reproduced verbatim below Each override was applied on its own and the plan re-solved, so every decision carries its own cost rather than one lumped figure. Overrides are never blocked. A veto removes an area, not a single lattice point, because a planner saying not there means a place.
---------------------------------	---

No overrides were applied to this run. The recommendation in Ex06 is the unmodified output of Ex05.

Ex07b – Safety gate decisions

SUPPORTS CAPTURED METHOD	That access, transfer and fallback decisions were taken deliberately Recorded at the point of each decision A gate never silently permits. Each row is a point at which the run either proceeded and recorded why that was safe, or refused. A refusal removes a source from the analysis and is stated here rather than left for the reader to infer from a missing figure.
---------------------------------	--

Gate	Outcome	Decided by	Reason recorded
source authorisation	PASS	harness	Water Point Data Exchange Plus (WPdx+) is a public endpoint; no authorisation required
source authorisation	PASS	harness	WorldPop gridded population is a public endpoint; no authorisation required
guardrail breach	PASS	harness	Overrides forgo 0 people, 0.0% of the achievable coverage, within the 5% tolerance set for this run.

Ex08 – Independent checks

SUPPORTS CAPTURED METHOD	That the plan was checked by a process that did not produce it Computed These checks run in a separate process from the one that produced the plan, can reject it, and do not read the override reasons in Ex07. They check the plan against the data, not against the account given of it. A single rejection stops the bundle being produced.
---------------------------------	---

Check	Result	Finding
coverage arithmetic	PASS	union recount matches at 299,034; across the 10 new sites a per-facility sum would report 34,700 against a true union of 34,700, an overcount of 0
geometry	PASS	all 10 sites lie inside the area of interest
budget	PASS	10 sites within the budget of 10
coordinate reference system	PASS	EPSG:32636 (WGS 84 / UTM zone 36N); over 400 sampled pairs the projected distance departs from great-circle by 0.290% at the median and 0.583% at most, or about 6 m on a 1000 m service radius
data currency	PASS	median source record is 4.5 years old

Check	Result	Finding
aggregation sensitivity	PASS	reweighting the demand surface within cells moves the coverage claim from 36.6% to 36.7%, a shift of 0.03%
boundary effect	FLAG	4 of 10 sites lie within one service radius of the district boundary; the nearest is 104 m from it. Population and facilities across the boundary are not represented, so coverage at these sites may be misstated
equity	FLAG	100% of newly covered people live in the densest quartile of cells, against 68% district-wide; the plan favours dense settlements over remote ones
provenance	PASS	4 sources recorded, all figures traceable
cartographic consistency	FLAG	the rendered figures were not reviewed against the account; run the map-reviewer agent over figures.json and pass its verdict with --figure-review

Ex08b – Scoring review

SUPPORTS CAPTURED METHOD	That the account was scored against a stated floor before issue Computed by a reviewer that reads only this bundle's results file The reviewer receives the results file and nothing else: not the solver, not the code path, not the override reasons. It cannot score the intention behind the plan, only the account given of it. Every dimension carries a floor and the weighted average carries a floor; the run is issued only when all are met, or when a person forces it and that is recorded.
--------------------------------	--

Dimension	Score	Floor	Weight		Basis
data adequacy	7.5	5.0	0.25	met	median record 4.5 years old; 1 semantic anomalies in the register
method fitness	6.5	5.0	0.20	met	greedy maximum covering with a stated 1 - 1/e bound; no exact benchmark (not run); coverage is straight-line, not travel time
spatial rigour	7.0	6.0	0.25	met	6 of 6 required diagnostics present; distances in EPSG:32636 (WGS 84 / UTM zone 36N)
accountability	9.0	6.0	0.20	met	every figure resolves to a recorded retrieval; 3 source anomalies disclosed
actionability	9.5	4.0	0.10	met	10 sites with coordinates and a stated basis; marginal coverage curve supports a budget decision; 3 sensitivity scenarios

Weighted score	7.67
Floor to issue	6.50
Decision	issue
Reviewer	deterministic rules over results.json

Ex09 – Sensitivity of the recommendation

SUPPORTS CAPTURED METHOD	How far the recommendation depends on assumptions the planner can change Computed Each scenario re-solves from the same data with one assumption changed. A recommendation that survives every row is one a planner can defend.
--------------------------------	---

Scenario	Covered	Share	Against the base case
Rehabilitation of existing points only	267,143	32.7%	-31,891
Half the budget (5 sites)	298,741	36.6%	-293
Double the budget (20 sites)	338,019	41.4%	+38,985

Ex10 – Anomalies in the source data, and how they were handled

SUPPORTS CAPTURED METHOD	That the source registers were read rather than assumed Observed during retrieval and cleaning Every item below is a property of the published data that would change a coverage figure if handled differently. Each is recorded with what was observed, what the agent did, and what a reader must keep in mind. None of these were corrected silently.
--------------------------------	--

1. SEMANTICS · WPdx+

Observed	status_clean is not two-valued in this district. Values present: Functional (1,174); Functional, needs repair (578); Non-Functional (241); Abandoned/Decommissioned (170); Functional, not in use (15).
Handling	Serving status is read from the status_semantics block in handbooks/wpdx.yaml, which treats Functional and Functional, needs repair as serving and everything else as not serving.
Bearing	An equality test against the string Functional would score every point in this district as not serving and inflate the gap.

2. CURRENCY · WPdx+

Observed	The median record in this district is 4.5 years old (oldest 16.3 years, newest 4.4 years).
Handling	Age is reported, not filtered. Uganda district holdings range from a mean of six to over sixteen years, so a staleness filter would delete whole districts rather than clean them.
Bearing	Coverage figures describe the surveyed state of the network, not necessarily its state today.

3. METHOD · MAP friction surface

Observed	Over this area the walking friction surface implies 14 minutes per kilometre at the median and 42 at the 90th percentile, so the chosen 1000 m corresponds to about 14 minutes of walking.
Handling	Coverage is tested as walking time rather than straight-line distance. Reach is computed per candidate from the local friction value rather than by a least-cost accumulation from every candidate, which would be exact but is not tractable at this candidate count.
Bearing	Where terrain changes sharply within one facility's reach, the local-friction approximation applies the conditions at the site in every direction, and is conservative rather than optimistic. The surface is modelled at about one kilometre and does not represent local footpaths.

Ex11 – Reproduction record

SUPPORTS	That this bundle can be regenerated from the recorded queries
CAPTURED	Computed
METHOD	The bundle is a function of one results file and the manifest that produced it. Any change to a source query, a cleaning rule or a retrieval date changes the provenance hash.

Run	masindi-water-20260827T030535Z
Generated	2026-08-27T03:06:57Z
Provenance hash	0e193d1063ade2ac

```
python -m siting.cli --country Uganda --adm2 Masindi --iso3 UGA --decisions decisions/masindi.yaml --format bundle
```

Source	Licence	Accessed
GADM database of Global Administrative Areas, version 4.1, https://gadm.org	GADM licence, free for academic and non-commercial use	2026-08-27
WorldPop, School of Geography and Environmental Science, University of Southampton, https://www.worldpop.org	CC BY 4.0	2026-08-27
Water Point Data Exchange (WPdx+), global water point registry, https://www.waterpointdata.org/	CC BY 4.0	2026-08-27